Farzaneh Barat

Ph.D. Student in Mechanical Engineering
Battery Modeling, Estimation, and Second-Life Systems
University of Kansas, Lawrence, KS, USA

barat@ku.edu
785-424-2521
Google Scholar

Education

University of Kansas — Lawrence, KS
Ph.D. in Mechanical Engineering — *Aug. 2023 – Present*

- **Advisor**: Huazhen Fang
- **Research Focus**: Lithium-ion battery modeling, parameter identification, electro-thermal analysis, health diagnostics, and second-life battery systems

University of Kashan — Kashan, Iran
M.Sc. in Mechanical Engineering — *Sep. 2015 – Feb. 2018*

- **Dissertation**: Solving path optimal tracking problem and determining load carrying capacity of manipulator using closed-loop combined optimization method

University of Kashan — Kashan, Iran
B.Sc. in Mechanical Engineering — *Sep. 2011 – Sep. 2015*

- **Final Project**: Hybrid solar–geothermal systems for heating and cooling of buildings

Work Experience

University of Kansas — Lawrence, KS
Graduate Research Assistant — *Aug. 2023 – Present*

- Research on lithium-ion battery modeling, parameter identification, electro-thermal analysis, and second-life battery systems.

University of Kansas — Lawrence, KS
Graduate Teaching Assistant — *Spring 2025*

- Supported laboratory sessions for LCA for Sustainable Design and Numerical Analysis, helping students with MATLAB, numerical methods, and engineering computation.

Diana Tajhiz Spadana Company and SEMAG Company — Isfahan, Iran
Head of R&D / Mechanical Designer — *Apr. 2018 – Jun. 2023*

- Led R&D and mechanical design projects for industrial product development.

University of Kashan — Kashan, Iran
Graduate Teaching Assistant & Instructor — *2015 – 2018*

Research Projects

- **Second-Life Lithium-Ion Battery Systems Modeling:** Investigating second-life lithium-ion battery modules through experimental testing, diagnostic characterization, performance analysis, and physics-informed modeling to evaluate degradation, safety, and reuse potential under practical operating and mechanical conditions for applications such as battery energy storage systems.
- **Recursive Parameter Identification of Nonlinear Stochastic State-Space Models:** Developed a sequentialized Ensemble Kalman Inversion framework for online parameter identification in nonlinear stochastic state-space models with latent states. The method combines particle-based state inference with Kalman-type parameter updates and was applied to lithium-ion battery model identification.

- **Parameter Identification of Battery Models:** Developing uncertainty-aware parameter identification frameworks for equivalent-circuit and nonlinear battery state-space models using Ensemble Kalman Inversion, with emphasis on accurate model calibration from voltage and temperature measurements.
- **Battery Health Diagnostics and State-of-Health (SOH)-Aware Modeling:** Extracting degradation features and developing SOH-aware models for lithium-ion batteries, with emphasis on reliable health estimation under demanding applications such as electric vertical takeoff and landing (eVTOL) systems.
- **Probabilistic Solver for PMP-Based Optimal Control:** Developing an uncertainty-aware probabilistic solver for continuous-time optimal control problems governed by Pontryagin's Minimum Principle, formulated as boundary-value problems and solved using probabilistic estimation and smoothing methods.

Publications

Conference Papers

- **F. Barat**, S. Wilson, H. Kim, and H. Fang, "System Identification of Lithium-Ion Battery Equivalent Circuit Models Using Ensemble Kalman Inversion," accepted for publication in the *Proceedings of the 2026 American Control Conference (ACC)*, 2026.
- **F. Barat** and M. Irani Rahaghi, "Solving nonlinear optimization problem using a new closed loop method for optimal tracking of a path," *2019 27th Iranian Conference on Electrical Engineering (ICEE)*, Yazd, Iran, pp. 1415–1420, May 2019. DOI: 10.1109/KBEI.2019.8734913

Journal Articles

- **F. Barat** and H. Fang, "System identification of nonlinear state-space models using Ensemble Kalman Inversion with application to lithium-ion batteries," under review at *IEEE Transactions on Control Systems Technology (TCST)*.
- M. Irani Rahaghi and **F. Barat**, "Solving nonlinear optimal path tracking problem using a new closed loop direct–indirect optimization method: Application on mechanical manipulators," *Robotica*, vol. 37, no. 1, pp. 39–61, Jan. 2019. DOI: 10.1017/S0263574718000851

Honors & Awards

- 3rd Place – KU Graduate Student Research Showcase, 2025
- Ranked 4th in M.Sc. Mechanical Engineering cohort, University of Kashan
- Ranked 1st in B.Sc. Heat and Fluid Mechanics, University of Kashan

Skills

- **Programming:** MATLAB/Simulink, Python
- **Engineering Software:** ANSYS, Abaqus, SolidWorks, Inventor, AutoCAD, CATIA
- **Hardware:** PEC battery tester
- **Tools:** LaTeX, Microsoft Office

References available upon request.